

Lab 2: Building a Reproducible Data Analysis Workflow

Purpose

You will practice working through a complete, reproducible data-analysis workflow in R. You will work with real psychological survey data and use R to clean, score, summarize, and communicate your findings.

Throughout the assignment, you will practice:

- cleaning and organizing raw survey data;
 - using `mutate()`, `select()`, and `filter()` to prepare data for analysis;
 - using `scoreItems()` and reverse-keying questionnaire items and calculate scale scores;
 - evaluating the internal consistency of psychological scales using Cronbach's alpha;
 - using `janitor` and `gt` to create reproducible tables;
 - using inline R code to automatically incorporate your results into a written report;
 - using Quarto to create a reproducible HTML document; and
 - using Git, GitHub, and GitHub Pages to save, share, and publish your analysis.
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Before You Begin

You must have the following installed.

- Quarto
- R **and** R Studio (you may need to use updated software for all steps to work)
- Have a GitHub account – you must input your name and email for this to function.
- You may also need to create and activate a GitHub token, see last week's materials for how to do this (in the Slack and on OWL).
- See here some documentation about how to use Quarto and what customizations you can make to chunks if you would like to learn more, or need to trouble shoot:
 - <https://quarto.org/docs/authoring/markdown-basics.html>
 - <https://quarto.org/docs/get-started/hello/rstudio.html>

Download the complete set of files provided for this lab.

If You Encounter a Problem

1. Are you trying to “do everything” in a Quarto document? Remember: it's sometimes easier to draft and debug code in an R script—the error messaging can be more straightforward, and there's less that can go wrong, as you're just focused on the R syntax, not yet also implementing it in Quarto

2. Inspect the error to see if it provides a solution.
3. Try to Google or ask AI agents what your error means, often technical errors can be solved quickly through these means.
4. Reach out to your classmates via Slack – it is likely that they have had a similar problem and can be of assistance.
5. Reach out to Selena via Slack

Remember that there is a learning curve with new software, and while frustration is common and real, practicing these workflows and problem solving will make this faster each time.

Part 1: Set Up the Project

1. Download the provided files.

Download the complete set of files provided for this lab.

These files constitute a folder which includes an already working, reproducible Quarto project, including a folder which includes:

- `bfi_index.qmd`: This is an example of a worked through assignment with a different dataset (Big Five Inventory)
- `bfi.xlsx`: This is the required dataset for this project

As well as the following:

- `README_Attachment and Social Anxiety.txt`: A text file with information regarding the scoring and labelling of items.
- `Attachment_Anxiety_Data.xlsx`: The data file you will use for your assignment
- `index.qmd`: an empty index qmd file for you to use for your assignment
- `_quarto.yml`: remember, this specialized file needs to be included in your directory for your GitHub Pages website to be created correctly. You don't have to do anything to it (though if you wish, you can update the title to something more informative)

Create a new Project file (.Rproj) for the lab (if using R Studio) and place the supplied files in the **top-level project directory**. You should not include the BFI files – this is simply meant to serve as an example.

- After creating a project, a folder on your computer will appear with the same name. You must place the given files (see Assignment Page), in this folder. They will then appear in your R Studio project page.

When creating the RStudio Project, check:

☒ **Create a git repository**

2. Create an empty GitHub repository

On GitHub, create a new repository using the naming convention specified.

lastname-lab-02

When creating it:

- do **not** add a README;
- do **not** add a `.gitignore`;
- do **not** add a license.

You want an empty remote repository.

3. Connect your local project to GitHub

Copy the URL of your new GitHub repository.

In the RStudio **Terminal**, run:

```
git remote add origin https://github.com/YOUR-USERNAME/YOUR-REPOSITORY.git
```

Verify the connection:

```
git remote -v
```

You should see your GitHub repository listed as `origin`.

Sometimes it is necessary for you to restart R Studio for the Git Pane (top right window) to become visible for committing, pushing, and pulling, etc.,

Part 2: Clean and Score Data

4. Load in packages and set up R chunks

Using `bfi_index.qmd` as a guide, load in required packages for cleaning data, and creating reproducible tables.

You must have all relevant packages installed. If you have never used R or these packages installed, please run the following in your R Source Panel

```
install.packages("tidyverse")
install.packages("psych")
install.packages("readxl")
install.packages("gt")
install.packages("janitor")
```

Recall that to begin an R chunk, you must type `````{r, CHUNK NAME} `````

Recall that using `#| output : FALSE` at the start of each chunk, prevents your raw code from rendering in your final reproducible document. See here some documentation about how to use Quarto and what customizations you can make to chunks:

<https://quarto.org/docs/authoring/markdown-basics.html>

<https://quarto.org/docs/get-started/hello/rstudio.html>

5. Load and inspect data

Import the `Attachment_Anxiety_Data.xlsx` (which should be in your R project directory at this point).

Now inspect your data. Notice that columns contain uninformative values. Open file `README_Attachment` and `Social Anxiety.txt` to investigate what meaningful values each numerical value corresponds to.

E.g., In the BFI dataset, under gender data values of 1 indicate Males and values of 2 indicate Females.

6. Clean and mutate data

Now, use the `mutate` function to assign informative labels to the dataset. You must do this for at least 2 of the 4 presented demographic variables (Gender, Age Group, Relationship Status, or Ethnicity).

Convert all survey response columns to object type **integer**.

7. Using the `list()` function, create an “Answer Key” which tells R which item responses belong to each Scale. You must do this for each scale SEPERATELY

Here it is important to note that you should reverse key items 5, 9, and 11 in the Social Interaction Anxiety Scale responses, and reverse key items 2, 5, 6, 8, and 9 in the Rosenberg Self-Esteem Scale.

8. Using the `scoreItems()` compute Cronbach’s alpha and average scores.

You must do this individually for each scale. Each scale has its own minimum and maximum value.

A. Social Interaction Anxiety Scale: minimum of 0, maximum of 4

B. Experiences in Close Relationships Anxiety Subscale: minimum of 1, maximum of 7

C. Rosenberg Self-Esteem Scale: minimum of 1, maximum of 4

Convert EACH new scores function into a dataframe

```
scores_scale.df <- as.data.frame(scores_scale$scores)
```

After computing easy score set, use `cbind()` to combine these into one dataframe.

E.g.

```
allscores.df <- cbind(scores_scale1.df, scores_scale2.df, scores_scale3.df)
```

9. Combine dataframes

Combine the dataframe you have now containing average scores, with your initial dataset containing all demographic variables.

10. Reduce dataframe

Using `select()` reduce this dataframe to only the columns which you are interested in.

Part 3: Create Reproducible Tables

11. Create a Demographic Table

Using the `tabyl()` function from package `janitor`, create a demographic summary table containing at least two demographic variables available in this dataset (these should be the ones which you used mutate on earlier).

```
demographic.table <- final.dataset |>
  tabyl(demographic_variable1, demographic_variable2)
```

E.g.

Using `gt()` turn this into a reproducible gt table with an intuitive title.

12. Create a Score Table

Using `describe()` capture summary statistics for the cleaned and scored dataset across all 3 scales. Assign this the name of "summary".

Extract and label the sample size using `describe()` or `nrow()`.

Create a column in your summary labelled "Scales".

Combine the three Cronbach's alpha values (one per scale), round them to 2 decimal places.

```
alphas <- round(c(
  scores_scale1$alpha,
  scores_scale2$alpha,
  scores_scale3$alpha
), 2)
```

Select only particular values from your summary table. This must include:

- Scale Titles
- Mean
- Median
- SD
- Range

Using `cbind()`, combine your summary table with your Cronbach alphas.

Using `gt()` turn this into a reproducible gt table with an intuitive title. Feel free to add subheadings as well as seen in `bfi_index.qmd`.

13. Filter your data.

Filter your cleaned data according to one demographic variable. For example, you might choose only participants of a specific relationship status, gender, or age group.

14. Create a FILTERED Score Table

Repeat step 12, now using your newly filtered dataset. Remember to use intuitive, but distinct object naming to ensure you do not rewrite your previously created objects. We want both tables (filtered and unfiltered) to exist at once.

Before putting anything on GitHub, make sure the provided workflow actually works on your computer.

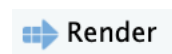
Part 4: Create your output document.

15. Set up your outline.

Using `##` or `###` for headings, create headings for “Intro”, “Methods”, “Participants”, “Measures”, “Data Analysis Strategy” and “Results”.

Ensure that this is done OUTSIDE of an R chunk.

Click render to ensure that this has worked.



Quarto should automatically create a `docs/` directory containing the rendered website.

Open the resulting webpage and inspect it.

Check:

- Does the document render without errors?
- Is there now a `docs/` directory in your project?

16. Fill in your sections with reproducible and editable data.

Under Participants

Insert inline R code which refers to your previously labelled sample size (Step 12).

This is done using ``{r}`` R CODE HERE`

Insert your first gt table from Step 11.

Under Measures

Describe the scales used in this assignment, using inline code to refer to their Cronbach's alphas from Step 8.

Under Data Analysis Strategy

Describe your reverse keying process.

Under Results

Insert your second gt table from step 12 using inline R code. Describe this table, inserting mean scores from step 12 for each scale using inline R code.

Insert your third gt table from step 14 using inline R code. Describe this table, inserting mean scores from step 14 for each scale using inline R code.

17. Save and Render

The project should render successfully.

18. Make your Git commit

Use the Git pane in RStudio to stage the project files (check the box next to each), including the generated `docs/` files.

Commit them with an informative message such as:

Final reproducible document.

Visit your GitHub repository and verify that the files are there.

If things won't pop up in GitHub. Type this into your terminal!

```
git remote add origin  
https://github.com/USERNAME/REPOSITORYNAME.git
```

```
git branch -M main
```

```
git push -u origin main
```

19. Configure GitHub Pages

On GitHub, navigate to:

Repository → **Settings** → **Pages**

Under **Build and deployment**, choose:

Source: Deploy from a branch

Branch: `main`

Folder: `/docs`

Save the settings.

GitHub should provide a URL resembling:

`https://YOUR-USERNAME.github.io/YOUR-REPOSITORY/`

Visit the URL.

Verify:

- Does the webpage load?
- Does it match the version you rendered locally?
- Can you find the source `index.qmd` in the GitHub repository?
- Can you also find the generated `docs/` files?

At this point you have a complete working pipeline:

local source → **rendered output** → **Git** → **GitHub** → **GitHub Pages**

14. Inspect the GitHub repository

Confirm that GitHub now contains:

- the repaired source code;
- the new filtering operation;
- the newly rendered `docs/` files; and
- only one step in the commit history

Your history should communicate a story something like:

What You Should Have Learned

By the end of this exercise, you should be able to:

- Upload and create an R Project
 - Create and populate R code chunks
 - Use dplyr functions like mutate(), select(), and filter()
 - Use scoreItems
 - Use describe() Local files
 - Create reproducible gt tables
 - Use inline code to create reproducible documents.
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Lab Completion Check

Confirm that you have:

- a working local RStudio/Git project;
- a GitHub repository connected to the project;
- a successfully rendered Quarto analysis;
- a working commit;
- a GitHub Pages site displaying the final analysis.

Upload a link to your GitHub Pages site as well as your GitHub Repository.